

1 Continuous-Time ESKF Error-State Derivation

1.1 Nominal state and IMU motion model

Define the nominal state

$$x = [p \quad v \quad R \quad b_g \quad b_a \quad g]^T.$$

The continuous-time IMU motion equations are

$$\dot{p} = v, \tag{1.1}$$

$$\dot{v} = R(\tilde{a} - b_a - \eta_a) + g, \tag{1.2}$$

$$\dot{R} = R(\tilde{\omega} - b_g - \eta_g)^\wedge, \tag{1.3}$$

$$\dot{b}_g = \eta_{bg}, \tag{1.4}$$

$$\dot{b}_a = \eta_{ba}, \tag{1.5}$$

$$\dot{g} = 0. \tag{1.6}$$

The IMU measurement equations may be written as

$$\tilde{a} = R^T(a - g) + b_a + \eta_a, \tag{1.7}$$

$$\tilde{\omega} = \omega + b_g + \eta_g, \tag{1.8}$$

together with

$$\dot{R} = R\omega^\wedge.$$

For the nominal state, the kinematic equations have the same form but ignore the measurement-noise terms:

$$\dot{p} = v, \tag{1.9}$$

$$\dot{v} = R(\tilde{a} - b_a) + g, \tag{1.10}$$

$$\dot{R} = R(\tilde{\omega} - b_g)^\wedge, \tag{1.11}$$

$$\dot{b}_g = 0, \tag{1.12}$$

$$\dot{b}_a = 0, \tag{1.13}$$

$$\dot{g} = 0. \tag{1.14}$$

1.2 True state

Define the true state as

$$x_t = [p_t \quad v_t \quad R_t \quad b_{at} \quad b_{gt} \quad g_t]^T.$$

Its continuous-time dynamics are

$$\dot{p}_t = v_t, \tag{1.15}$$

$$\dot{v}_t = R_t(\tilde{a} - b_{at} - \eta_a) + g_t, \tag{1.16}$$

$$\dot{R}_t = R_t(\tilde{\omega} - b_{gt} - \eta_g)^\wedge, \tag{1.17}$$

$$\dot{b}_{gt} = \eta_{bg}, \tag{1.18}$$

$$\dot{b}_{at} = \eta_{ba}, \tag{1.19}$$

$$\dot{g}_t = 0. \tag{1.20}$$

1.3 Definition of the error state

Use a right-multiplicative rotation error:

$$p_t = p + \delta p, \quad (1.21)$$

$$v_t = v + \delta v, \quad (1.22)$$

$$R_t = R \delta R = R \text{Exp}(\delta \theta), \quad (1.23)$$

$$b_{gt} = b_g + \delta b_g, \quad (1.24)$$

$$b_{at} = b_a + \delta b_a, \quad (1.25)$$

$$g_t = g + \delta g. \quad (1.26)$$

The quantities without the subscript t are nominal-state variables. The goal is to derive the continuous-time dynamics of

$$\delta x = [\delta p \quad \delta v \quad \delta \theta \quad \delta b_g \quad \delta b_a \quad \delta g]^T.$$

1.4 Position error

Starting from

$$p_t = p + \delta p,$$

differentiate with respect to time:

$$\dot{p}_t = \dot{p} + \delta \dot{p}.$$

Since

$$\dot{p}_t = v_t = v + \delta v$$

and

$$\dot{p} = v,$$

we obtain

$$v + \delta v = v + \delta \dot{p}.$$

Therefore

$$\boxed{\delta \dot{p} = \delta v.} \quad (1.27)$$

1.5 Velocity error

Start from

$$v_t = v + \delta v.$$

Differentiating,

$$\dot{v}_t = \dot{v} + \delta \dot{v}.$$

Using the true-state equation,

$$\dot{v}_t = R_t(\tilde{a} - b_{at} - \eta_a) + g_t.$$

Substitute the error-state definitions:

$$\dot{v}_t = R \text{Exp}(\delta \theta)(\tilde{a} - b_a - \delta b_a - \eta_a) + g + \delta g.$$

For small $\delta \theta$,

$$\text{Exp}(\delta \theta) = I + \delta \theta^\wedge + O(\|\delta \theta\|^2).$$

Keeping only first-order error terms,

$$\dot{v}_t \approx R(I + \delta\theta^\wedge)(\tilde{a} - b_a - \delta b_a - \eta_a) + g + \delta g \quad (1.28)$$

$$\approx R\tilde{a} - Rb_a - R\delta b_a - R\eta_a + R\delta\theta^\wedge\tilde{a} - R\delta\theta^\wedge b_a + g + \delta g. \quad (1.29)$$

Terms such as

$$\delta\theta^\wedge\delta b_a, \quad \delta\theta^\wedge\eta_a$$

are neglected as higher-order terms in this first-order error-state linearization.

Using

$$a^\wedge b = -b^\wedge a,$$

we have

$$\delta\theta^\wedge\tilde{a} = -\tilde{a}^\wedge\delta\theta, \quad -\delta\theta^\wedge b_a = b_a^\wedge\delta\theta.$$

Hence

$$\dot{v}_t \approx R(\tilde{a} - b_a) + g - R(\tilde{a} - b_a)^\wedge\delta\theta - R\delta b_a - R\eta_a + \delta g.$$

On the other hand, the nominal equation gives

$$\dot{v} + \delta\dot{v} = R(\tilde{a} - b_a) + g + \delta\dot{v}.$$

Comparing the two expressions,

$$\boxed{\delta\dot{v} = -R(\tilde{a} - b_a)^\wedge\delta\theta - R\delta b_a - \eta_a + \delta g.} \quad (1.30)$$

Here we use the standard ESKF noise assumption adopted in the text: multiplying the zero-mean white accelerometer noise by a rotation does not change its distribution. Thus, in distribution,

$$R\eta_a \sim \eta_a,$$

and the rotated noise term is written simply as

$$R\eta_a \Rightarrow \eta_a.$$

This is a distributional simplification rather than a pointwise identity between individual noise samples.

1.6 Rotation error

Start from

$$R_t = R \text{Exp}(\delta\theta).$$

Differentiating,

$$\dot{R}_t = \dot{R} \text{Exp}(\delta\theta) + R \frac{d}{dt} \text{Exp}(\delta\theta).$$

Exact derivative of the exponential map

For a general time-varying

$$\delta\theta(t) \in \mathbb{R}^3,$$

the exact derivative is

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge}, \quad (1.31)$$

where $J_r(\delta\theta)$ is the right Jacobian of $\text{SO}(3)$.

The formula above can be seen directly from the limit definition of the right Jacobian.

Define

$$Q(t) = \text{Exp}(\delta\theta(t)).$$

The right/local tangent velocity of $Q(t)$ is

$$\omega_{\text{local}} = \lim_{h \rightarrow 0} \frac{\text{Log}(Q(t)^{-1} Q(t+h))}{h}.$$

Since

$$Q(t)^{-1} = \text{Exp}(-\delta\theta(t))$$

and

$$Q(t+h) = \text{Exp}(\delta\theta(t+h)),$$

we have

$$\omega_{\text{local}} = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-\delta\theta(t)) \text{Exp}(\delta\theta(t+h))]}{h}.$$

Using the ordinary first-order expansion

$$\delta\theta(t+h) = \delta\theta(t) + h \delta\dot{\theta}(t) + O(h^2),$$

this becomes

$$\omega_{\text{local}} = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-\delta\theta) \text{Exp}(\delta\theta + h \delta\dot{\theta})]}{h}.$$

Now compare this with the directional-limit definition of the right Jacobian:

$$\boxed{J_r(\phi)u = \lim_{h \rightarrow 0} \frac{\text{Log}[\text{Exp}(-\phi) \text{Exp}(\phi + hu)]}{h}}. \quad (1.32)$$

By identifying

$$\phi = \delta\theta, \quad u = \delta\dot{\theta},$$

we obtain

$$\boxed{\omega_{\text{local}} = J_r(\delta\theta) \delta\dot{\theta}}. \quad (1.33)$$

Since a right/local tangent velocity satisfies

$$\dot{Q} = Q \omega_{\text{local}}^\wedge,$$

we get

$$\dot{Q} = Q \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge.$$

Substituting

$$Q = \text{Exp}(\delta\theta)$$

gives the exact relation

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge} . \quad (1.34)$$

Equivalently, this is the same relation as the compact right-Jacobian increment formula

$$\text{Exp}(\delta\theta + \delta\dot{\theta} dt) \approx \text{Exp}(\delta\theta) \text{Exp}\left(J_r(\delta\theta) \delta\dot{\theta} dt \right) .$$

To see why these two forms are equivalent, define

$$Q(t) = \text{Exp}(\delta\theta(t)) .$$

Starting from the differential form,

$$\dot{Q} = Q \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge ,$$

the ordinary first-order Taylor expansion in time gives

$$Q(t + dt) = Q(t) + \dot{Q}(t) dt + O(dt^2) .$$

Substituting the differential equation,

$$Q(t + dt) = Q(t) + Q(t) \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge dt + O(dt^2) .$$

Factoring out $Q(t)$,

$$Q(t + dt) = Q(t) \left[I + \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge dt \right] + O(dt^2) .$$

For a small vector ϵ ,

$$\text{Exp}(\epsilon) = I + \epsilon^\wedge + O(\|\epsilon\|^2) .$$

Choose

$$\epsilon = J_r(\delta\theta) \delta\dot{\theta} dt .$$

Then

$$\text{Exp}\left(J_r(\delta\theta) \delta\dot{\theta} dt \right) = I + \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge dt + O(dt^2) .$$

Therefore

$$Q(t + dt) \approx Q(t) \text{Exp}\left(J_r(\delta\theta) \delta\dot{\theta} dt \right) .$$

Since

$$Q(t) = \text{Exp}(\delta\theta(t))$$

and

$$\delta\theta(t + dt) = \delta\theta(t) + \delta\dot{\theta}(t) dt + O(dt^2) ,$$

we obtain

$$\boxed{\text{Exp}(\delta\theta + \delta\dot{\theta} dt) \approx \text{Exp}(\delta\theta) \text{Exp}\left(J_r(\delta\theta) \delta\dot{\theta} dt \right)} . \quad (1.35)$$

Conversely, start from the increment form:

$$Q(t + dt) \approx Q(t) \text{Exp}\left(J_r(\delta\theta)\delta\dot{\theta} dt\right).$$

Using the small-angle expansion of the second exponential,

$$Q(t + dt) = Q(t) + Q(t) \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge dt + O(dt^2).$$

Hence

$$\frac{Q(t + dt) - Q(t)}{dt} = Q(t) \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge + O(dt).$$

Taking the limit $dt \rightarrow 0$ gives

$$\boxed{\dot{Q} = Q \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge.} \quad (1.36)$$

Thus the increment form and the differential form describe the same local motion:

$$\boxed{\begin{aligned} \text{Exp}(\delta\theta + \delta\dot{\theta} dt) &\approx \text{Exp}(\delta\theta) \text{Exp}\left(J_r(\delta\theta)\delta\dot{\theta} dt\right) \\ &\Updownarrow dt \rightarrow 0 \\ \frac{d}{dt} \text{Exp}(\delta\theta) &= \text{Exp}(\delta\theta) \left(J_r(\delta\theta)\delta\dot{\theta}\right)^\wedge. \end{aligned}} \quad (1.37)$$

The upper expression is a first-order finite-increment relation, while the lower expression is the corresponding exact instantaneous differential relation.

Since

$$J_r(\delta\theta) = I - \frac{1}{2}\delta\theta^\wedge + O(\|\delta\theta\|^2),$$

the first-order ESKF approximation is

$$J_r(\delta\theta)\delta\dot{\theta} \approx \delta\dot{\theta}.$$

Therefore, within the first-order error-state derivation,

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) \approx \text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge.} \quad (1.38)$$

Using this first-order approximation,

$$\dot{R}_t \approx \dot{R} \text{Exp}(\delta\theta) + R \text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge.$$

The nominal rotation dynamics are

$$\dot{R} = R(\tilde{\omega} - b_g)^\wedge.$$

Hence

$$\dot{R}_t \approx R(\tilde{\omega} - b_g)^\wedge \text{Exp}(\delta\theta) + R \text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge.$$

The true rotation dynamics are

$$\dot{R}_t = R_t(\tilde{\omega} - b_{gt} - \eta_g)^\wedge = R \text{Exp}(\delta\theta)(\tilde{\omega} - b_{gt} - \eta_g)^\wedge.$$

Equating both expressions and canceling R gives

$$\text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge = \text{Exp}(\delta\theta)(\tilde{\omega} - b_{gt} - \eta_g)^\wedge \quad (1.39)$$

$$- (\tilde{\omega} - b_g)^\wedge \text{Exp}(\delta\theta). \quad (1.40)$$

Use the $\text{SO}(3)$ identity

$$\phi^\wedge R = R(R^T \phi)^\wedge.$$

With

$$R = \text{Exp}(\delta\theta),$$

this gives

$$(\tilde{\omega} - b_g)^\wedge \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) (\text{Exp}(-\delta\theta)(\tilde{\omega} - b_g))^\wedge.$$

Therefore

$$\text{Exp}(\delta\theta)\delta\dot{\theta}^\wedge = \text{Exp}(\delta\theta) \left[(\tilde{\omega} - b_{gt} - \eta_g)^\wedge \right. \quad (1.41)$$

$$\left. - (\text{Exp}(-\delta\theta)(\tilde{\omega} - b_g))^\wedge \right]. \quad (1.42)$$

For small $\delta\theta$,

$$\text{Exp}(-\delta\theta) \approx I - \delta\theta^\wedge.$$

Thus

$$(\tilde{\omega} - b_{gt} - \eta_g) - \text{Exp}(-\delta\theta)(\tilde{\omega} - b_g) \quad (1.43)$$

$$\approx (\tilde{\omega} - b_{gt} - \eta_g) - (I - \delta\theta^\wedge)(\tilde{\omega} - b_g) \quad (1.44)$$

$$= b_g - b_{gt} - \eta_g + \delta\theta^\wedge \tilde{\omega} - \delta\theta^\wedge b_g. \quad (1.45)$$

Since

$$b_{gt} = b_g + \delta b_g,$$

we have

$$b_g - b_{gt} = -\delta b_g.$$

Also,

$$\delta\theta^\wedge(\tilde{\omega} - b_g) = -(\tilde{\omega} - b_g)^\wedge \delta\theta.$$

Therefore

$$\delta\dot{\theta} = -(\tilde{\omega} - b_g)^\wedge \delta\theta - \delta b_g - \eta_g.$$

Hence

$$\boxed{\delta\dot{\theta} = -(\tilde{\omega} - b_g)^\wedge \delta\theta - \delta b_g - \eta_g.} \quad (1.46)$$

Identity used above

For $R \in \text{SO}(3)$,

$$\text{Ad}_R \phi = (R\phi^\wedge R^T)^\vee = R\phi.$$

Thus

$$R\phi^\wedge R^T = (R\phi)^\wedge.$$

From this,

$$\phi^\wedge R = R(R^T \phi)^\wedge.$$

1.7 Supplement: From Finite Infinitesimal Increments to Instantaneous Velocity

This appendix clarifies how the right-Jacobian finite-increment relation is used to obtain the instantaneous right/local tangent velocity.

Let

$$Q(t) = \text{Exp}(\delta\theta(t)).$$

For a small time increment dt ,

$$\delta\theta(t + dt) = \delta\theta(t) + \delta\dot{\theta}(t) dt + O(dt^2),$$

and therefore

$$Q(t + dt) = \text{Exp}(\delta\theta(t + dt)) \approx \text{Exp}(\delta\theta + \delta\dot{\theta} dt).$$

The right-Jacobian increment relation is

$$\boxed{\text{Exp}(\phi + \Delta\phi) \approx \text{Exp}(\phi) \text{Exp}(J_r(\phi)\Delta\phi).} \quad (1.47)$$

Set

$$\phi = \delta\theta, \quad \Delta\phi = \delta\dot{\theta} dt.$$

Then

$$\boxed{Q(t + dt) \approx Q(t) \text{Exp}\left(J_r(\delta\theta)\delta\dot{\theta} dt\right).} \quad (1.48)$$

Now define the instantaneous right/local tangent velocity ω_R by the infinitesimal motion

$$\boxed{Q(t + dt) \approx Q(t) \text{Exp}(\omega_R dt).} \quad (1.49)$$

Comparing the two infinitesimal expressions gives

$$\boxed{\omega_R = J_r(\delta\theta)\delta\dot{\theta}.} \quad (1.50)$$

Thus $\delta\dot{\theta}$ is the time derivative of the exponential coordinates, whereas $J_r(\delta\theta)\delta\dot{\theta}$ is the corresponding right/local tangent velocity of the rotation matrix.

To recover the ordinary matrix derivative, use

$$\text{Exp}(\omega_R dt) = I + \omega_R^\wedge dt + O(dt^2).$$

Hence

$$Q(t + dt) = Q(t) + Q(t)\omega_R^\wedge dt + O(dt^2),$$

so

$$\frac{Q(t + dt) - Q(t)}{dt} = Q(t)\omega_R^\wedge + O(dt).$$

Taking $dt \rightarrow 0$,

$$\boxed{\dot{Q} = Q\omega_R^\wedge.} \quad (1.51)$$

Substituting

$$\omega_R = J_r(\delta\theta)\delta\dot{\theta}$$

and

$$Q = \text{Exp}(\delta\theta)$$

gives

$$\boxed{\frac{d}{dt} \text{Exp}(\delta\theta) = \text{Exp}(\delta\theta) \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge.} \quad (1.52)$$

Therefore the finite infinitesimal increment and the instantaneous differential form are two descriptions of the same local motion:

$$\boxed{\begin{aligned} Q(t+dt) &\approx Q(t) \text{Exp}\left(J_r(\delta\theta) \delta\dot{\theta} dt\right) \\ &\Downarrow \quad dt \rightarrow 0 \\ \dot{Q} &= Q \left(J_r(\delta\theta) \delta\dot{\theta} \right)^\wedge. \end{aligned}} \quad (1.53)$$

Exact and small-angle forms of the right Jacobian

For $\phi \in \mathbb{R}^3$, let

$$\vartheta = \|\phi\|.$$

The exact right Jacobian of $\text{SO}(3)$ is

$$\boxed{J_r(\phi) = I - \frac{1 - \cos \vartheta}{\vartheta^2} \phi^\wedge + \frac{\vartheta - \sin \vartheta}{\vartheta^3} (\phi^\wedge)^2.} \quad (1.54)$$

The coefficient of the first-order term follows from

$$\cos \vartheta = 1 - \frac{\vartheta^2}{2!} + \frac{\vartheta^4}{4!} - \cdots,$$

so

$$\frac{1 - \cos \vartheta}{\vartheta^2} = \frac{1}{2} - \frac{\vartheta^2}{24} + \cdots.$$

Similarly,

$$\sin \vartheta = \vartheta - \frac{\vartheta^3}{3!} + \frac{\vartheta^5}{5!} - \cdots,$$

which gives

$$\frac{\vartheta - \sin \vartheta}{\vartheta^3} = \frac{1}{6} - \frac{\vartheta^2}{120} + \cdots.$$

Therefore

$$J_r(\phi) = I - \frac{1}{2} \phi^\wedge + \frac{1}{6} (\phi^\wedge)^2 + O(\|\phi\|^3),$$

and, retaining only first-order terms,

$$\boxed{J_r(\phi) = I - \frac{1}{2} \phi^\wedge + O(\|\phi\|^2).} \quad (1.55)$$

For the ESKF rotation error, $\phi = \delta\theta$ is small. Hence

$$J_r(\delta\theta) \delta\dot{\theta} = \delta\dot{\theta} - \frac{1}{2} \delta\theta^\wedge \delta\dot{\theta} + \cdots.$$

Under the first-order error-state approximation, the product involving $\delta\theta$ and $\delta\dot{\theta}$ is neglected, yielding

$$\boxed{J_r(\delta\theta) \delta\dot{\theta} \approx \delta\dot{\theta}.} \quad (1.56)$$

This is why the ESKF derivation commonly uses

$$\frac{d}{dt} \text{Exp}(\delta\theta) \approx \text{Exp}(\delta\theta) \delta\dot{\theta}^\wedge,$$

even though the exact differential relation contains the right Jacobian.

1.8 Gyroscope-bias error

Starting from

$$b_{gt} = b_g + \delta b_g,$$

differentiate:

$$\dot{b}_{gt} = \dot{b}_g + \delta \dot{b}_g.$$

The true bias dynamics are

$$\dot{b}_{gt} = \eta_{bg},$$

while the nominal model ignores noise:

$$\dot{b}_g = 0.$$

Therefore

$$\boxed{\delta \dot{b}_g = \eta_{bg}.} \tag{1.57}$$

1.9 Accelerometer-bias error

Similarly,

$$b_{at} = b_a + \delta b_a.$$

Differentiating,

$$\dot{b}_{at} = \dot{b}_a + \delta \dot{b}_a.$$

Using

$$\dot{b}_{at} = \eta_{ba}, \quad \dot{b}_a = 0,$$

we obtain

$$\boxed{\delta \dot{b}_a = \eta_{ba}.} \tag{1.58}$$

1.10 Gravity error

Starting from

$$g_t = g + \delta g,$$

differentiate:

$$\dot{g}_t = \dot{g} + \delta \dot{g}.$$

Since

$$\dot{g}_t = 0, \quad \dot{g} = 0,$$

we obtain

$$\boxed{\delta \dot{g} = 0.} \tag{1.59}$$

2 Final continuous-time error-state equations

Collecting the results,

$$\begin{aligned}
 \delta \dot{p} &= \delta v, \\
 \delta \dot{v} &= -R(\tilde{a} - b_a)^\wedge \delta \theta - R\delta b_a - \eta_a + \delta g, \\
 \delta \dot{\theta} &= -(\tilde{\omega} - b_g)^\wedge \delta \theta - \delta b_g - \eta_g, \\
 \delta \dot{b}_g &= \eta_{bg}, \\
 \delta \dot{b}_a &= \eta_{ba}, \\
 \delta \dot{g} &= 0.
 \end{aligned} \tag{2.1}$$

3 Discrete-Time ESKF Kinematic Equation

Notation convention

In the continuous-time derivations below, t , τ , and s denote physical time. After discretization, the final propagation equations use the discrete time-step indices k and $k+1$, with

$$t_k = t, \quad t_{k+1} = t_k + \Delta t = t + \Delta t. \tag{3.1}$$

Thus, continuous-time integration uses t, τ, s , while final discrete-time propagation equations use $k, k+1$.

3.1 Nominal-state position equation

For the nominal state,

$$\dot{p} = v, \quad \dot{v} = R(\tilde{a} - b_a) + g.$$

Define the acceleration expressed in the world frame as

$$a_W(t) = R(t)(\tilde{a}(t) - b_a(t)) + g. \tag{3.2}$$

Since

$$\dot{p}(\tau) = v(\tau),$$

integrating from t to $t + \Delta t$ gives

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} v(\tau) d\tau.$$

The velocity at an intermediate time τ is

$$v(\tau) = v(t) + \int_t^\tau a_W(s) ds.$$

Substituting this into the position integral,

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} \left[v(t) + \int_t^\tau a_W(s) ds \right] d\tau.$$

Assumption over one IMU sampling interval

To obtain the discrete-time equation used here, assume that the world-frame acceleration is approximately constant over the short IMU sampling interval:

$$\boxed{a_W(s) \approx a_W(t), \quad s \in [t, t + \Delta t].} \quad (3.3)$$

Equivalently, over this interval we use the values at the beginning of the interval,

$$R(s) \approx R(t), \quad \tilde{a}(s) \approx \tilde{a}(t), \quad b_a(s) \approx b_a(t),$$

with gravity g treated as constant. Thus

$$a_W(s) \approx R(t)(\tilde{a}(t) - b_a(t)) + g.$$

Under this approximation,

$$\int_t^\tau a_W(s) ds \approx a_W(t)(\tau - t).$$

Therefore

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} [v(t) + a_W(t)(\tau - t)] d\tau \quad (3.4)$$

$$= p(t) + v(t)\Delta t + a_W(t) \int_t^{t+\Delta t} (\tau - t) d\tau. \quad (3.5)$$

The remaining integral is

$$\int_t^{t+\Delta t} (\tau - t) d\tau = \frac{1}{2}\Delta t^2.$$

Hence

$$p(t + \Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}a_W(t)\Delta t^2.$$

Substituting the expression for $a_W(t)$,

$$p(t + \Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}[R(t)(\tilde{a}(t) - b_a(t)) + g]\Delta t^2.$$

Finally,

$$\boxed{p_{k+1} = p_k + v_k\Delta t + \frac{1}{2}R_k(\tilde{a}_k - b_{a,k})\Delta t^2 + \frac{1}{2}g_k\Delta t^2.} \quad (3.6)$$

This is the discrete-time nominal position propagation equation. The appearance of $R(t)$ rather than $R(t + \Delta t)$ follows from the beginning-of-interval approximation above. In the exact continuous motion, $R(s)$ and therefore $a_W(s)$ generally vary throughout the sampling interval.

3.2 Nominal-state velocity equation

Starting from

$$\dot{v} = R(\tilde{a} - b_a) + g \equiv a_W,$$

integrate over the interval $[t, t + \Delta t]$:

$$v(t + \Delta t) = v(t) + \int_t^{t+\Delta t} a_W(\tau) d\tau.$$

Using the same assumption as in the position derivation,

$$a_W(\tau) \approx a_W(t), \quad \tau \in [t, t + \Delta t],$$

we obtain

$$v(t + \Delta t) \approx v(t) + a_W(t) \int_t^{t+\Delta t} d\tau.$$

Therefore

$$v(t + \Delta t) = v(t) + a_W(t) \Delta t.$$

Substituting

$$a_W(t) = R(t)(\tilde{a}(t) - b_a(t)) + g,$$

gives

$$\boxed{v_{k+1} = v_k + R_k(\tilde{a}_k - b_{a,k})\Delta t + g_k\Delta t.} \quad (3.7)$$

This is the discrete-time nominal velocity propagation equation. As with the position equation, the approximation uses the beginning-of- interval values $R(t)$, $\tilde{a}(t)$, and $b_a(t)$ over one IMU sampling interval.

3.3 Nominal-state rotation equation

The continuous-time nominal rotation dynamics are

$$\boxed{\dot{R}(t) = R(t)(\tilde{\omega}(t) - b_g(t))^\wedge.} \quad (3.8)$$

Define the bias-corrected body-frame angular velocity

$$\omega_B(t) \equiv \tilde{\omega}(t) - b_g(t).$$

Over one short IMU sampling interval, assume

$$\boxed{\omega_B(\tau) \approx \omega_B(t), \quad \tau \in [t, t + \Delta t].} \quad (3.9)$$

Equivalently,

$$\tilde{\omega}(\tau) \approx \tilde{\omega}(t), \quad b_g(\tau) \approx b_g(t)$$

over this interval. The differential equation therefore becomes

$$\dot{R}(\tau) = R(\tau)\omega_B(t)^\wedge,$$

where $\omega_B(t)^\wedge$ is constant with respect to τ .

The solution of this matrix differential equation with initial condition $R(t)$ is

$$R(\tau) = R(t) \exp(\omega_B(t)^\wedge(\tau - t)),$$

where $\exp(\cdot)$ denotes the matrix exponential.

This can be verified directly:

$$\frac{d}{d\tau} R(\tau) = R(t) \frac{d}{d\tau} \exp(\omega_B(t)^\wedge(\tau - t)) \quad (3.10)$$

$$= R(t) \exp(\omega_B(t)^\wedge(\tau - t)) \omega_B(t)^\wedge \quad (3.11)$$

$$= R(\tau) \omega_B(t)^\wedge. \quad (3.12)$$

At $\tau = t + \Delta t$,

$$R(t + \Delta t) = R(t) \exp(\omega_B(t)^\wedge \Delta t) .$$

Since

$$\text{Exp}(\phi) \equiv \exp(\phi^\wedge)$$

and

$$(\omega_B \Delta t)^\wedge = \omega_B^\wedge \Delta t,$$

we obtain

$$R(t + \Delta t) = R(t) \text{Exp}(\omega_B(t) \Delta t) .$$

Substituting

$$\omega_B(t) = \tilde{\omega}(t) - b_g(t),$$

gives the discrete-time nominal rotation propagation equation

$$\boxed{R_{k+1} = R_k \text{Exp}((\tilde{\omega}_k - b_{g,k}) \Delta t)} . \quad (3.13)$$

The exponential increment appears on the right because

$$\dot{R} = R \omega_B^\wedge$$

uses the body-frame (right-trivialized) angular velocity. In contrast, if the angular velocity were expressed in the world frame so that

$$\dot{R} = \omega_W^\wedge R,$$

the corresponding constant-angular-velocity propagation would be

$$R(t + \Delta t) = \text{Exp}(\omega_W \Delta t) R(t).$$

3.4 Nominal-state bias and gravity equations

For the nominal state, the gyroscope bias, accelerometer bias, and gravity are modeled as constant during propagation:

$$\dot{b}_g = 0, \quad \dot{b}_a = 0, \quad \dot{g} = 0.$$

Integrating the gyroscope-bias equation over one sampling interval,

$$b_g(t + \Delta t) = b_g(t) + \int_t^{t+\Delta t} \dot{b}_g(\tau) d\tau,$$

gives

$$\boxed{b_{g,k+1} = b_{g,k}} . \quad (3.14)$$

Similarly,

$$b_a(t + \Delta t) = b_a(t) + \int_t^{t+\Delta t} \dot{b}_a(\tau) d\tau,$$

so

$$\boxed{b_{a,k+1} = b_{a,k}} . \quad (3.15)$$

For gravity,

$$g(t + \Delta t) = g(t) + \int_t^{t+\Delta t} \dot{g}(\tau) d\tau,$$

and therefore

$$\boxed{g_{k+1} = g_k.} \quad (3.16)$$

The constant nominal-bias equations do not imply that the physical IMU biases are assumed to remain exactly constant. Their random-walk variations are represented in the error-state dynamics by

$$\delta \dot{b}_g = \eta_{bg}, \quad \delta \dot{b}_a = \eta_{ba},$$

and their uncertainty is propagated through the error-state covariance.

3.5 Summary of the discrete-time nominal-state equations

Collecting the nominal-state propagation equations,

$$\boxed{\begin{aligned} p_{k+1} &= p_k + v_k \Delta t + \frac{1}{2} [R_k(\tilde{a}_k - b_{a,k}) + g_k] \Delta t^2, \\ v_{k+1} &= v_k + [R_k(\tilde{a}_k - b_{a,k}) + g_k] \Delta t, \\ R_{k+1} &= R_k \text{Exp}((\tilde{\omega}_k - b_{g,k}) \Delta t), \\ b_{g,k+1} &= b_{g,k}, \\ b_{a,k+1} &= b_{a,k}, \\ g_{k+1} &= g_k. \end{aligned}} \quad (3.17)$$

3.6 Discrete-time error-state equations

We now discretize the continuous-time error-state equations derived previously.

Position error

From the continuous-time position-error equation,

$$\delta \dot{p} = \delta v,$$

assume that δv is approximately constant over one short sampling interval $[t, t + \Delta t]$. Then

$$\delta p(t + \Delta t) - \delta p(t) = \int_t^{t+\Delta t} \delta \dot{p}(\tau) d\tau \quad (3.18)$$

$$= \int_t^{t+\Delta t} \delta v(\tau) d\tau \quad (3.19)$$

$$\approx \delta v(t) \Delta t. \quad (3.20)$$

Therefore,

$$\boxed{\delta p_{k+1} = \delta p_k + \delta v_k \Delta t.} \quad (3.21)$$

Velocity error

From the continuous-time velocity-error equation,

$$\delta \dot{v} = -R(\tilde{a} - b_a)^\wedge \delta \theta - R \delta b_a - \eta_a + \delta g,$$

define the deterministic part

$$\boxed{f_v = -R(\tilde{a} - b_a)^\wedge \delta\theta - R\delta b_a + \delta g.} \quad (3.22)$$

Then

$$\delta\dot{v} = f_v - \eta_a.$$

Integrating over one sampling interval,

$$\delta v(t + \Delta t) - \delta v(t) = \int_t^{t+\Delta t} f_v(\tau) d\tau - \int_t^{t+\Delta t} \eta_a(\tau) d\tau. \quad (3.23)$$

Assume that the deterministic quantities in f_v are approximately constant over the short interval:

$$\boxed{f_v(\tau) \approx f_v(t), \quad \tau \in [t, t + \Delta t].} \quad (3.24)$$

Thus

$$\int_t^{t+\Delta t} f_v(\tau) d\tau \approx f_v(t)\Delta t.$$

Define the integrated accelerometer-noise increment

$$\boxed{\eta_v \equiv \int_t^{t+\Delta t} \eta_a(\tau) d\tau.} \quad (3.25)$$

Assume the continuous-time accelerometer noise is zero-mean white Gaussian noise with

$$\mathbb{E}[\eta_a(t)] = 0$$

and

$$\boxed{\mathbb{E}[\eta_a(t)\eta_a(s)^T] = \sigma_a^2 I \delta(t - s).} \quad (3.26)$$

Equivalently, σ_a^2 is the continuous-time white-noise intensity (per axis). Then the integrated noise increment also has zero mean:

$$\mathbb{E}[\eta_v] = 0.$$

Its covariance is

$$\text{Cov}(\eta_v) = \mathbb{E}[\eta_v \eta_v^T] \quad (3.27)$$

$$= \int_t^{t+\Delta t} \int_t^{t+\Delta t} \mathbb{E}[\eta_a(\tau)\eta_a(s)^T] d\tau ds \quad (3.28)$$

$$= \sigma_a^2 I \int_t^{t+\Delta t} \int_t^{t+\Delta t} \delta(\tau - s) d\tau ds \quad (3.29)$$

$$= \boxed{\sigma_a^2 \Delta t I.} \quad (3.30)$$

Thus, per axis,

$$\boxed{\text{Var}(\eta_v) = \sigma_a^2 \Delta t.} \quad (3.31)$$

The book distinguishes the sampled accelerometer measurement noise from the continuous-time noise-density parameter through

$$\boxed{\sigma_a(k) = \frac{\sigma_a}{\sqrt{\Delta t}.} \quad (3.32)$$

Therefore the standard deviation of the integrated velocity-noise increment can be written in either equivalent form:

$$\boxed{\sigma(\eta_v) = \Delta t \sigma_a(k) = \sigma_a \sqrt{\Delta t}.} \quad (3.33)$$

The first expression uses the sampled-noise notation, while the second follows directly after substituting the sampled-noise relation.

Note that writing simply

$$\text{Var}(\eta_a) = \sigma_a^2$$

for continuous-time white noise is shorthand for its noise intensity. The Dirac-delta covariance above makes the continuous-to-discrete scaling explicit.

Hence

$$\delta v(t + \Delta t) \approx \delta v(t) + f_v(t)\Delta t - \eta_v.$$

Substituting the expression for $f_v(t)$ gives

$$\boxed{\begin{aligned} \delta v(t + \Delta t) = & \delta v(t) - R(t)(\tilde{a}(t) - b_a(t))^\wedge \delta \theta(t) \Delta t \\ & - R(t) \delta b_a(t) \Delta t + \delta g(t) \Delta t - \eta_v. \end{aligned}} \quad (3.34)$$

The term η_v is the discrete-time noise increment obtained by integrating the continuous-time accelerometer noise over the sampling interval; unlike the deterministic term $f_v(t)\Delta t$, it should not be interpreted simply as a deterministic constant multiplied by Δt .

Rotation error

From the continuous-time rotation-error equation,

$$\delta \dot{\theta} = -(\tilde{\omega} - b_g)^\wedge \delta \theta - \delta b_g - \eta_g,$$

define

$$A \equiv -(\tilde{\omega} - b_g)^\wedge.$$

Over one short IMU sampling interval, assume that

$$\tilde{\omega}(\tau) \approx \tilde{\omega}(t), \quad b_g(\tau) \approx b_g(t),$$

so that A is approximately constant. The homogeneous equation

$$\dot{x} = Ax$$

has the solution

$$x(t + \Delta t) = \exp(A\Delta t)x(t).$$

Therefore the homogeneous part of the rotation error propagates as

$$\delta \theta(t + \Delta t) = \exp(-(\tilde{\omega}(t) - b_g(t))^\wedge \Delta t) \delta \theta(t).$$

Using the SO(3) exponential-map notation,

$$\exp(-(\tilde{\omega} - b_g)^\wedge \Delta t) = \text{Exp}(-(\tilde{\omega} - b_g)\Delta t).$$

For the forcing terms $-\delta b_g - \eta_g$, the first-order discretization used here gives

$$\boxed{\delta\theta(t + \Delta t) \approx \text{Exp}(-(\tilde{\omega}(t) - b_g(t))\Delta t) \delta\theta(t) - \delta b_g(t)\Delta t - \eta_\theta,} \quad (3.35)$$

where the integrated gyroscope-noise increment is

$$\boxed{\eta_\theta \equiv \int_t^{t+\Delta t} \eta_g(\tau) d\tau.} \quad (3.36)$$

More exactly, for constant

$$A = -(\tilde{\omega} - b_g)^\wedge,$$

the variation-of-constants formula is

$$\delta\theta(t + \Delta t) = e^{A\Delta t} \delta\theta(t) - \int_t^{t+\Delta t} e^{A(t+\Delta t-\tau)} [\delta b_g(\tau) + \eta_g(\tau)] d\tau.$$

The expression above with $-\delta b_g\Delta t - \eta_\theta$ is the corresponding first-order forcing-term approximation.

Variance of the discrete rotation-noise increment. Assume the continuous-time gyroscope noise is zero-mean isotropic white Gaussian noise with

$$\mathbb{E}[\eta_g(t)] = 0$$

and

$$\boxed{\mathbb{E}[\eta_g(t)\eta_g(s)^T] = \sigma_g^2 I \delta(t - s).} \quad (3.37)$$

Then

$$\mathbb{E}[\eta_\theta] = 0$$

and

$$\text{Cov}(\eta_\theta) = \mathbb{E}[\eta_\theta \eta_\theta^T] \quad (3.38)$$

$$= \int_t^{t+\Delta t} \int_t^{t+\Delta t} \mathbb{E}[\eta_g(\tau)\eta_g(s)^T] d\tau ds \quad (3.39)$$

$$= \boxed{\sigma_g^2 \Delta t I.} \quad (3.40)$$

Thus, per axis,

$$\boxed{\text{Var}(\eta_\theta) = \sigma_g^2 \Delta t.} \quad (3.41)$$

For the sampled gyroscope measurement noise, the book uses

$$\boxed{\sigma_g(k) = \frac{\sigma_g}{\sqrt{\Delta t}}.} \quad (3.42)$$

Hence the standard deviation of the integrated rotation-noise increment can be written as

$$\boxed{\sigma(\eta_\theta) = \Delta t \sigma_g(k) = \sigma_g \sqrt{\Delta t}.} \quad (3.43)$$

Again, the first form uses the sampled-noise notation, while the second is the equivalent expression in terms of the continuous-time noise-density parameter. The $\sqrt{\Delta t}$ factor belongs to the standard deviation, not to the variance. The same covariance result also holds if the exact state-transition rotation inside the noise integral is retained: because A is skew-symmetric, e^{As} is a rotation matrix, and for isotropic noise

$$e^{As}(\sigma_g^2 I)e^{AsT} = \sigma_g^2 I.$$

Gyroscope-bias error

Here we explicitly distinguish the continuous-time bias random-walk process from its integrated discrete-time increment. The continuous process is denoted by

$$\eta_{bg}(t),$$

while

$$\eta_{bg,d}$$

denotes the noise accumulated over one sampling interval. The book reuses the symbol η_{bg} for the discrete increment in Eqs. (3.42)–(3.43); the subscript d is retained here to avoid ambiguity.

From

$$\delta \dot{b}_g = \eta_{bg},$$

integrating over one sampling interval gives

$$\delta b_g(t + \Delta t) = \delta b_g(t) + \int_t^{t+\Delta t} \eta_{bg}(\tau) d\tau.$$

Define

$$\boxed{\eta_{bg,d} \equiv \int_t^{t+\Delta t} \eta_{bg}(\tau) d\tau.} \quad (3.44)$$

Then

$$\boxed{\delta b_{g,k+1} = \delta b_{g,k} + \eta_{bg,d}.} \quad (3.45)$$

If

$$\mathbb{E}[\eta_{bg}(t)\eta_{bg}(s)^T] = \sigma_{bg}^2 I \delta(t-s),$$

then

$$\boxed{\text{Cov}(\eta_{bg,d}) = \sigma_{bg}^2 \Delta t I,} \quad (3.46)$$

or, per axis,

$$\boxed{\text{Var}(\eta_{bg,d}) = \sigma_{bg}^2 \Delta t, \quad \sigma_{\eta_{bg,d}} = \sigma_{bg} \sqrt{\Delta t}.} \quad (3.47)$$

Accelerometer-bias error

Likewise, $\eta_{ba}(t)$ denotes the continuous-time accelerometer-bias random-walk process, while $\eta_{ba,d}$ denotes its integrated discrete-time increment over one sampling interval.

Similarly, from

$$\delta \dot{b}_a = \eta_{ba},$$

we obtain

$$\boxed{\delta b_a(t + \Delta t) = \delta b_a(t) + \eta_{ba,d},} \quad (3.48)$$

where

$$\boxed{\eta_{ba,d} \equiv \int_t^{t+\Delta t} \eta_{ba}(\tau) d\tau.} \quad (3.49)$$

For continuous-time isotropic white noise intensity σ_{ba}^2 ,

$$\boxed{\text{Cov}(\eta_{ba,d}) = \sigma_{ba}^2 \Delta t I,} \quad (3.50)$$

and therefore

$$\boxed{\text{Var}(\eta_{ba,d}) = \sigma_{ba}^2 \Delta t, \quad \sigma_{\eta_{ba,d}} = \sigma_{ba} \sqrt{\Delta t}.} \quad (3.51)$$

Gravity error

From

$$\delta \dot{g} = 0,$$

integration gives

$$\delta g(t + \Delta t) - \delta g(t) = \int_t^{t+\Delta t} \delta \dot{g}(\tau) d\tau = 0.$$

Therefore

$$\boxed{\delta g_{k+1} = \delta g_k.} \quad (3.52)$$

3.7 Summary of the discrete-time error-state equations

Collecting the discrete-time error-state propagation equations,

$$\boxed{\begin{aligned} \delta p_{k+1} &= \delta p_k + \delta v_k \Delta t, \\ \delta v_{k+1} &= \delta v_k - R_k(\tilde{a}_k - b_{a,k})^\wedge \delta \theta_k \Delta t - R_k \delta b_{a,k} \Delta t + \delta g_k \Delta t - \eta_v, \\ \delta \theta_{k+1} &\approx \text{Exp}(-(\tilde{\omega}_k - b_{g,k}) \Delta t) \delta \theta_k - \delta b_{g,k} \Delta t - \eta_\theta, \\ \delta b_{g,k+1} &= \delta b_{g,k} + \eta_{bg,d}, \\ \delta b_{a,k+1} &= \delta b_{a,k} + \eta_{ba,d}, \\ \delta g_{k+1} &= \delta g_k. \end{aligned}} \quad (3.53)$$

It is useful to list both the standard deviations and variances. The sampled IMU measurement-noise standard deviations satisfy

$$\sigma_a(k) = \frac{\sigma_a}{\sqrt{\Delta t}}, \quad \sigma_g(k) = \frac{\sigma_g}{\sqrt{\Delta t}}.$$

Therefore,

$$\boxed{\begin{aligned} \sigma(\eta_v) &= \Delta t \sigma_a(k) = \sigma_a \sqrt{\Delta t}, & \text{Var}(\eta_v) &= \sigma_a^2 \Delta t, \\ \sigma(\eta_\theta) &= \Delta t \sigma_g(k) = \sigma_g \sqrt{\Delta t}, & \text{Var}(\eta_\theta) &= \sigma_g^2 \Delta t, \\ \sigma(\eta_{bg,d}) &= \sigma_{bg} \sqrt{\Delta t}, & \text{Var}(\eta_{bg,d}) &= \sigma_{bg}^2 \Delta t, \\ \sigma(\eta_{ba,d}) &= \sigma_{ba} \sqrt{\Delta t}, & \text{Var}(\eta_{ba,d}) &= \sigma_{ba}^2 \Delta t. \end{aligned}} \quad (3.54)$$

The first two standard-deviation expressions use the sampled IMU-noise notation. For the bias random walks, the book denotes the integrated discrete noises again by η_{bg} and η_{ba} ; this note uses $\eta_{bg,d}$ and $\eta_{ba,d}$ so that they remain distinct from the continuous-time processes $\eta_{bg}(t)$ and $\eta_{ba}(t)$.

4 Motion Process of ESKF

Using the discrete-time error-state equations derived above, write the error-state motion model as

$$\boxed{\delta x_{k+1} = f(\delta x_k) + w_k, \quad w_k \sim \mathcal{N}(0, Q_k).} \quad (4.1)$$

The discrete process-noise covariance is

$$\boxed{Q_k = \text{diag}(0_3, \text{Cov}(\eta_v), \text{Cov}(\eta_\theta), \text{Cov}(\eta_{bg,d}), \text{Cov}(\eta_{ba,d}), 0_3).} \quad (4.2)$$

Here the subscript d is retained for the bias-noise increments in order to distinguish them from the corresponding continuous-time white-noise processes.

4.1 Linearized form

Linearize the discrete transition function f about the nominal error-state mean. For a general linearization point $\bar{\delta x}_k$,

$$f(\delta x_k) \approx f(\bar{\delta x}_k) + \left. \frac{\partial f(\delta x)}{\partial \delta x} \right|_{\delta x = \bar{\delta x}_k} (\delta x_k - \bar{\delta x}_k).$$

In the ESKF, the error-state mean is reset to zero, so the linearization point is

$$\bar{\delta x}_k = 0.$$

Therefore,

$$f(\delta x_k) \approx f(0) + \left. \frac{\partial f(\delta x)}{\partial \delta x} \right|_{\delta x = 0} \delta x_k.$$

For the error-state motion model,

$$f(0) = 0.$$

Define the discrete-time error-state transition Jacobian as

$$F_k \triangleq \left. \frac{\partial f(\delta x)}{\partial \delta x} \right|_{\delta x = 0}. \quad (4.3)$$

The linearized error-state propagation is therefore

$$\delta x_{k+1} \approx F_k \delta x_k + w_k. \quad (4.4)$$

Thus, F_k is the Jacobian of the *discrete-time error-state transition function* with respect to the error state, evaluated at zero error. It should not be confused with the time derivative $d\delta x/dt$.

4.2 State transition Jacobian matrix

The error-state ordering is

$$\delta x = [\delta p \quad \delta v \quad \delta \theta \quad \delta b_g \quad \delta b_a \quad \delta g]^T. \quad (4.5)$$

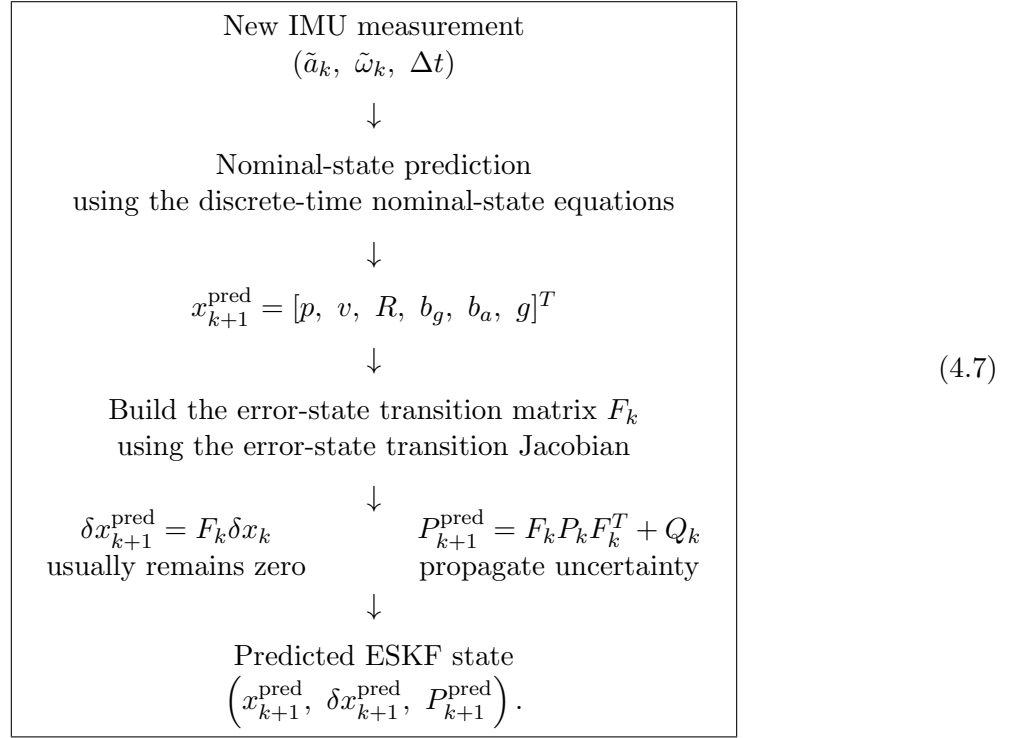
Since the discrete-time error-state equations have already been linearized explicitly, the transition Jacobian can be read directly from the coefficients multiplying each error-state component. With the ordering above,

$$F_k = \begin{bmatrix} I_3 & I_3 \Delta t & 0 & 0 & 0 & 0 \\ 0 & I_3 & -R(\tilde{a} - b_a)^\wedge \Delta t & 0 & -R \Delta t & I_3 \Delta t \\ 0 & 0 & \text{Exp}(-(\tilde{\omega} - b_g) \Delta t) & -I_3 \Delta t & 0 & 0 \\ 0 & 0 & 0 & I_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_3 \end{bmatrix}. \quad (4.6)$$

Each entry is a 3×3 block, so $F_k \in \mathbb{R}^{18 \times 18}$.

4.3 ESKF IMU propagation flow

For each new IMU measurement, the ESKF prediction process can be summarized as follows:



Mathematically, the prediction contains three steps:

1. propagate the nominal state using the IMU measurements;
2. propagate the error-state mean,

$$\delta x_{k+1}^{\text{pred}} = F_k \delta x_k;$$

3. propagate the error-state covariance,

$$P_{k+1}^{\text{pred}} = F_k P_k F_k^T + Q_k.$$

In the ESKF, the error-state mean is reset to zero after the update step. Therefore, in the usual prediction cycle,

$$\delta x_k = 0 \implies \delta x_{k+1}^{\text{pred}} = F_k \delta x_k = 0.$$

Consequently, the explicit error-state mean propagation is often unnecessary in an implementation. The important quantities propagated during the IMU prediction are the nominal state and the error-state covariance.

5 Update Process of ESKF

5.1 Observation model

Consider the general observation equation

$z = h(x) + v, \quad v \sim \mathcal{N}(0, V).$

(5.1)

Here z is the observation data, v is the observation noise, and V is the observation-noise covariance matrix.

In a traditional EKF, the observation equation is linearized with respect to the state variables. In the ESKF, the Kalman-filter update is performed with respect to the error-state variables. Therefore, the required observation Jacobian is

$$H = \left. \frac{\partial h}{\partial \delta x} \right|_{x_{\text{pred}}} . \quad (5.2)$$

5.2 Kalman update equations

Once the observation Jacobian is available, the Kalman gain is

$$K = P_{\text{pred}} H^T (H P_{\text{pred}} H^T + V)^{-1} . \quad (5.3)$$

The error-state update is

$$\delta x = K (z - h(x_{\text{pred}})) . \quad (5.4)$$

Following the notation used in this chapter, the nominal state is then updated as

$$x = x_{\text{pred}} + \delta x . \quad (5.5)$$

The covariance update is

$$P = (I - KH) P_{\text{pred}} . \quad (5.6)$$

Here K is the Kalman gain, P_{pred} is the predicted error-state covariance, and P is the updated covariance.

5.3 Observation Jacobian with respect to the error state

For observations of the nominal state, the Jacobian with respect to the error state can be obtained using the chain rule:

$$H = \frac{\partial h}{\partial x} \frac{\partial x}{\partial \delta x} . \quad (5.7)$$

The chain of mappings is

$$\delta x \longrightarrow x(\delta x) \longrightarrow h(x(\delta x)) .$$

Here $x(\delta x)$ denotes a local perturbation of the nominal state used for linearization. It should not be confused with the true state.

5.4 Local perturbation parameterization

For the nominal-state ordering

$$x = [p \quad v \quad R \quad b_g \quad b_a \quad g]^T ,$$

define the locally perturbed state by

$$x(\delta x) = \begin{bmatrix} p + \delta p \\ v + \delta v \\ R \text{Exp}(\delta \theta) \\ b_g + \delta b_g \\ b_a + \delta b_a \\ g + \delta g \end{bmatrix}, \quad x(0) = x_{\text{pred}}. \quad (5.8)$$

For the Euclidean components,

$$\frac{\partial(p + \delta p)}{\partial \delta p} = I_3, \quad \frac{\partial(v + \delta v)}{\partial \delta v} = I_3,$$

and similarly

$$\frac{\partial(b_g + \delta b_g)}{\partial \delta b_g} = I_3, \quad \frac{\partial(b_a + \delta b_a)}{\partial \delta b_a} = I_3, \quad \frac{\partial(g + \delta g)}{\partial \delta g} = I_3.$$

The rotation component requires special treatment. Since the perturbation is applied on the right,

$$R(\delta \theta) = R \text{Exp}(\delta \theta),$$

and the corresponding rotation coordinate is

$$\text{Log}(R \text{Exp}(\delta \theta)).$$

Therefore,

$$\left. \frac{\partial \text{Log}(R \text{Exp}(\delta \theta))}{\partial \delta \theta} \right|_{\delta \theta=0} = J_r^{-1}(\text{Log } R). \quad (5.9)$$

Consequently,

$$\frac{\partial x}{\partial \delta x} = \text{diag} (I_3, I_3, J_r^{-1}(\text{Log } R), I_3, I_3, I_3). \quad (5.10)$$

5.5 Full-pose GNSS observation

For the full-pose GNSS case, the observation contains position and rotation. With the nominal-state ordering above, the observation function is conceptually

$$h(x) = \begin{bmatrix} p \\ R \end{bmatrix}. \quad (5.11)$$

Thus the GNSS observation directly depends on the nominal position p and nominal rotation R , while it does not directly observe v , b_g , b_a , or g .

The next step is to evaluate

$$\frac{\partial h}{\partial x} = \begin{bmatrix} \frac{\partial h}{\partial p} & \frac{\partial h}{\partial v} & \frac{\partial h}{\partial R} & \frac{\partial h}{\partial b_g} & \frac{\partial h}{\partial b_a} & \frac{\partial h}{\partial g} \end{bmatrix},$$

and then combine it with $\partial x / \partial \delta x$. The treatment of the rotation observation and its relation to the direct error-state GNSS formulation is left for the following discussion.